CS & IT ENGINEERING



Database Management System

Query Languages

DPP 01 Discussion Notes



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[MCQ]

#Q. Consider the following statements:

S_1: $\pi_{\text{List N}}\left(\pi_{\text{List N}-1}\ldots\ldots\left(\pi_{\text{List 1}}(R)\right)\right)$ — Commutative

$\equiv \pi_{\text{List 1}}\left(\pi_{\text{List 2}}\ldots\ldots\left(\pi_{\text{List N}}(R)\right)\right)$

S_2: $\sigma_{c_n}\left(\sigma_{c_{n-1}}\ldots\ldots\left(\sigma_{c_1}(R)\right)\right) \equiv \sigma_{c_1}\left(\sigma_{c_2}\ldots\left(\sigma_{C_N}(C)\right)\right)$

Which of the following statement(s) is/are correct?

A. S_1 only

B. S_2 only

C. Both S_1 and S_2 only

D. Neither S_1 nor S_2

[MCQ]

#Q. Consider the following table

1 + 4

I(pq)

p	q
0	1
4	5
8	9

J(qr)

q	r
1	2
5	2
5	6
5	10
13	10

K(rs)

r	s
2	3
6	7
10	11
10	3

The number of tuples in $(I \bowtie J \bowtie K)$ where $\bowtie$ is the natural join is

A 5

B 8

C 10

D 11

[MCQ]

#Q. Consider the following relations:

Enroll (Sid, Papercode), Paper(Papercode, Desc)

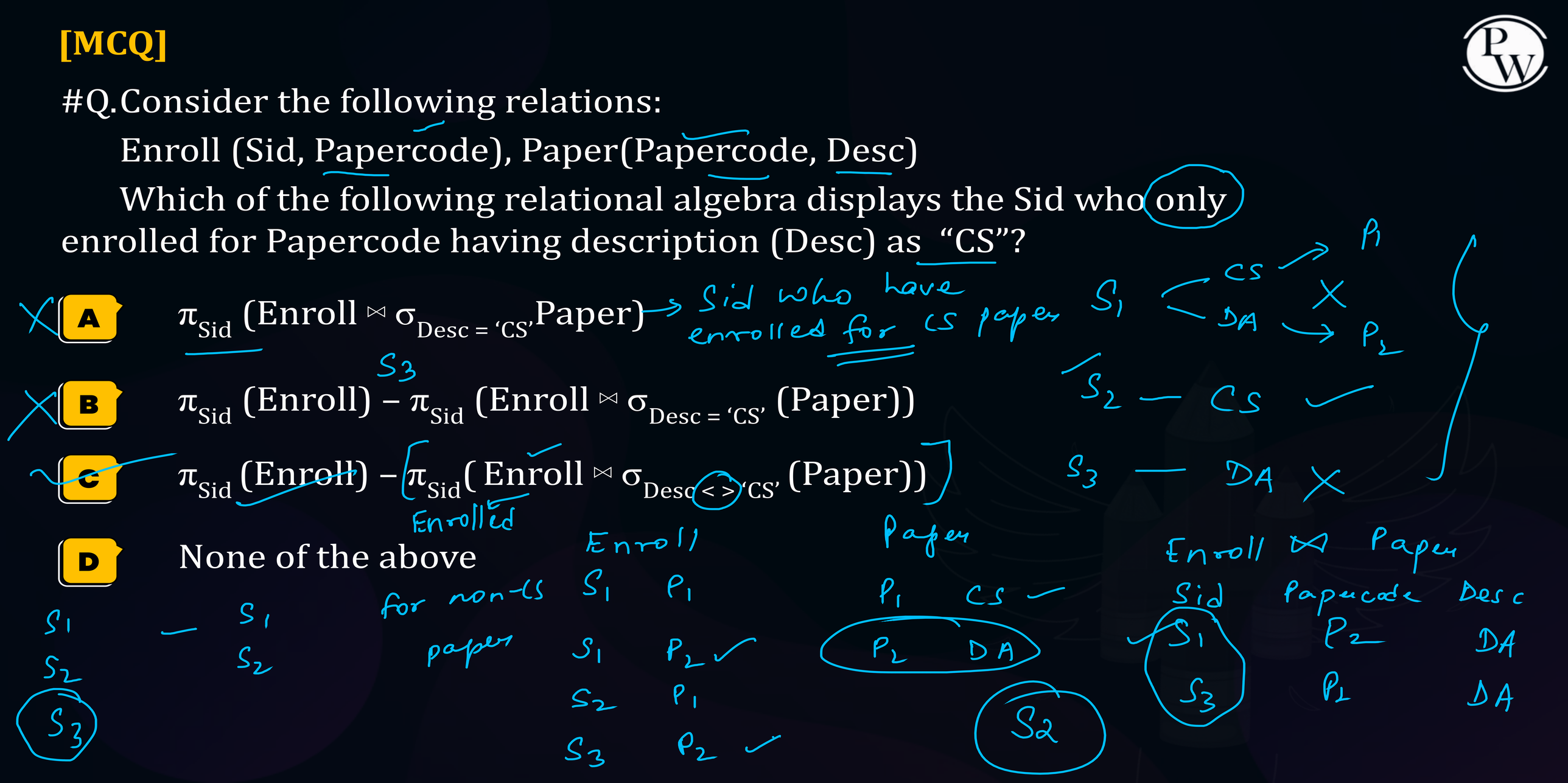
Which of the following relational algebra displays the Sid who only enrolled for Papercode having description (Desc) as "CS"?

A π_{Sid} (Enroll $\bowtie$ $\sigma_{Desc = 'CS'}$ Paper)

B π_{Sid} (Enroll) – π_{Sid} (Enroll $\bowtie$ $\sigma_{Desc = 'CS'}$ (Paper))

C π_{Sid} (Enroll) – [π_{Sid} (Enroll $\bowtie$ $\sigma_{Desc <> 'CS'}$ (Paper))]

D None of the above



[MCQ]

#Q.Consider a relation, work (EmpID, ProjectID)

The suitable relational algebra expression that projects the employee ids who work in exactly one project is

A $\pi_{EmpID}(work) - \pi_{EmpID}(work \bowtie_{(EmpID = E \ \wedge \ ProjectID = P)} \rho_{E,P}(work))$

B $\pi_{EmpID}(work \bowtie_{(EmpID = E \ \wedge \ ProjectID \neq P)} \rho_{E,P}(work))$

C $\pi_{EmpID}(work) - \pi_{EmpID}(work \bowtie_{(EmpID = E \ \wedge \ ProjectID \neq P)} \rho_{E,P}(work))$

D None of the above

[MCQ]

#Q. Consider the relation R(ABCD) which of the following relational algebra expression return the lowest value of B? (R_1 and R_2 are rename of R)

I. $\Pi_{R_2 \cdot B}\ (R_1 \bowtie_{R1.B < R2.B} R_2)$

II. $\Pi_B(R) - \Pi_{R_1 \cdot B}\ (R_1 \bowtie_{R1.B > R2.B} R_2)$

III. $\Pi_B(R) - \Pi_{R_1 \cdot B}\ (R_1 \bowtie_{R1.B < R2.B} R_2)$

IV. $\Pi_B(R)$

R(A B C D)

A	B	C	D
1	2	3	4
5	6	1	8
2	1	9	10
⋮			

A I

B II

C III

D IV

[MCQ]

#Q. Consider the following RA expression-

P : π_{sid} (student) – π_{sid} (student $\bowtie_{Marks < M \wedge Gender = G}$ $\rho_{(I, G, M)}$ (student))

on a relation student(sid, Gender, Marks)

The above query displays

S_1	M	80
S_2	F	65
S_3	F	90
S_4	M	72

A. The sid of the students who obtained the maximum marks.

B. The sids of the male and female students who obtained the maximum marks in their respective gender.

C. The sids of male students who scored higher than all the female students

D. None of the above

[MSQ]

#Q. Consider the relation-

Works (Eid , Pid), Project (Pid, Name)

The relational algebra expression that displays the Eids who work in every project with Name = 'M' is

Eid is not working in that project Pid in M

A $\pi_{Eid, Pid}$ (Works) / π_{Pid} ($\sigma_{Name = 'M'}$ (Project))

B π_{Eid} (Works) - π_{Eid}[(π_{Eid} (Works) × π_{Pid} ($\sigma_{Name = 'M'}$ (Project))) - $\pi_{Eid, Pid}$ (Works)]

C π_{Eid} (Works) - π_{Eid}[(π_{Eid} (Works) × π_{Pid} ($\sigma_{Name <> 'M'}$ (Project))) - $\pi_{Eid, Pid}$ (Works)]

D None of the above

[MCQ]

#Q.Consider the two relations R_1 and R_2 such that they have no attributes in common then-

S_1: $R_1 \bowtie R_2 = R_1 \times R_2$

S_2: $R_1 \bowtie R_2 = \varphi$

Which of the following is correct?

$R_1(A, B)$ $R_2(C, D)$

A S_1 only

B S_2 only

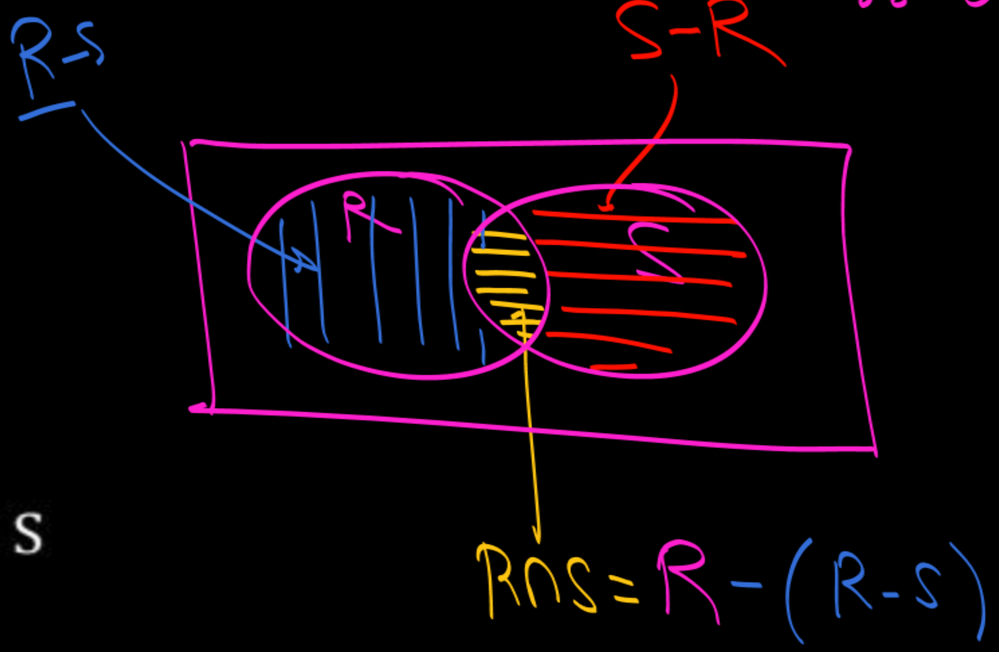
C Both S_1 and S_2 only

D Neither S_1 nor S_2

[MSQ] [Mark-2]

#Q. Suppose that two relations R (A, B) and S (A, B) have exactly the same schema. Which of the following is/are always true

∴ Union Compatible



(A) $R \cap S = R - (R - S)$

(B) $R \cap S = R - (S - R)$

(C) $R \cap S = R$ NATURAL JOIN S

(D) $R \cap S = \sigma_{R.A=S.A \wedge R.B=S.B}(R \times S)$

attributes = 2

attributes = 4

[MSQ] [Mark-2]

#Q. Consider two relations R(A,B) and S(B,C) and the following relational algebra expression S: $\pi_{R.A,\ S.B}(\sigma_{R.B=S.B}\ (R \times S))$
Which of the following relational algebra expressions are guaranteed to produce same result as S

A $\pi_{A,B}\ (R \bowtie S)$

B $R \bowtie \pi_B\ (S)$

C $R \cap (\pi_A\ (R) \times \pi_B\ (S))$

D $\pi_{A,\ R.B}\ (R \times S)$

[MCQ] -2 Mark

S_1	P_2	30
S_3	P_3	30

S_1, S_3

Catalog

S_1	P_2	30
S_3	P_3	30

#Q. Consider the following relational schemas

Supplier(Sid, Sname), Parts(Pid, Pname), Catalog(Sid, Pid, Cost)

and following two queries

- Q1: π_{Sid} [Catalog − [$\pi_{C1.Sid, C1.Pid, C1.cost}(\sigma_{(C1.cost < C2.cost)}(\rho_{C1}(Catalog) \times \rho_{C2}(Catalog))$]]
- Q2: π_{Sid} [Catalog] − $\pi_{C1.Sid}$ [$\pi_{C1.Sid, C1.Pid, C1.cost}(\sigma_{(C1.cost < C2.cost)}(\rho_{C1}(Catalog) \times \rho_{C2}(Catalog))$]

$\{S_1, S_3\} - \{S_1\} = \{S_3\}$

A. Both Q1 and Q2 always produces the same output.

B. Output produced by Q1 is always different for output produced by Q2.

C. Output produced by Q1 is subset of output produced by Q2.

D. Output produced by Q2 is subset of output produced by Q1.

THANK - YOU

